

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (5 points) Evaluate the integral using the indicated substitution.

$$\int \frac{4x}{\sqrt{2x^2+5}} dx, \quad u = 2x^2 + 5$$

Solution: Take $u = 2x^2 + 5$, then $du = (4x)dx$.

$$\begin{aligned} \int \frac{4x}{\sqrt{2x^2+5}} dx &= \int \frac{du}{\sqrt{u}} \\ &= 2\sqrt{u} + C \\ &= 2\sqrt{2x^2+5} + C \end{aligned}$$

Problem 2. (5 points) Evaluate the integral using an appropriate substitution.

$$\int \sin^4 3x \cos 3x dx$$

Solution: Take $u = \sin 3x$, then $du = (3 \cos 3x) dx$.

$$\begin{aligned} \int \sin^4 3x \cos 3x dx &= \frac{1}{3} \int \sin^4 3x \cdot (3 \cos 3x) dx \\ &= \frac{1}{3} \int u^4 du \\ &= \frac{1}{15} u^5 + C \\ &= \frac{1}{15} \sin^5 3x + C \end{aligned}$$